Unit 7, Lesson 8: Using Trigonometric Ratios and the Pythagorean Theorem

Benchmark

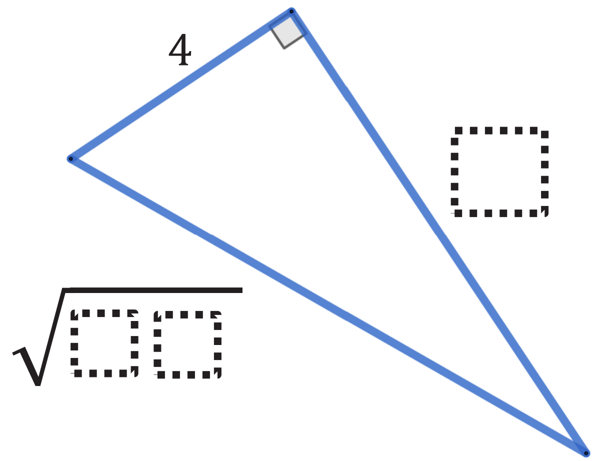
MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Learning Targets

- ☐ I can use trigonometric ratios to solve mathematical problems.
- ☐ I can use the Pythagorean Theorem to solve mathematical problems.

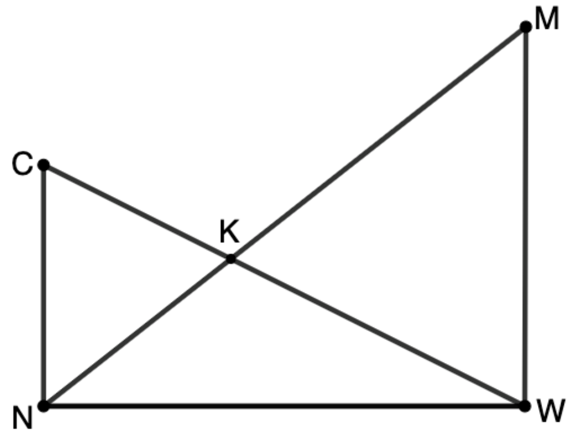
7.8.1 Warm-Up: Critical Thinking

1. Use the digits 0-9 one time each to find the missing side lengths of the right triangle. The triangle is not drawn to scale.



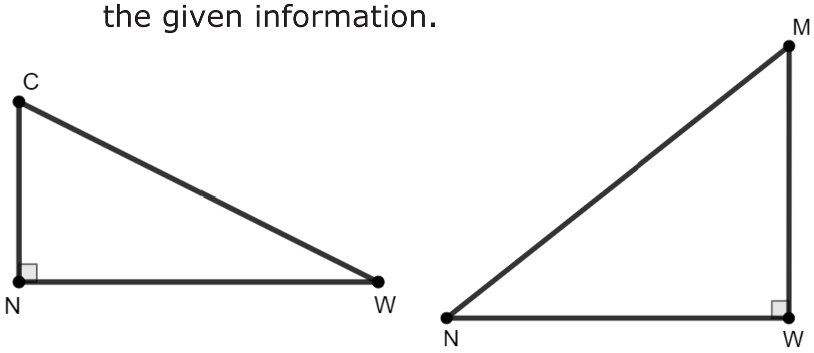
7.8.2 Collaboration: Overlapping Triangles

1. ΔMWN and ΔCNW are right triangles that share side $\overline{NW}$ and K is the intersection point of sides $\overline{CW}$ and $\overline{MN}$. In the triangles, the $m\angle MWN = m\angle CNW = 90°$, $MW = 4.4$, $NW = 6$, and $m\angle NCK = 63°$. ΔCNK is an isosceles triangle with base $\overline{CK}$ and ΔMWK is an isosceles triangle with base $\overline{WK}$.



a. Mark the figure with all information given in the problem.

b. Jasmine redrew the diagram as two separate triangles, as shown. Mark Jasmine's diagram using the given information.



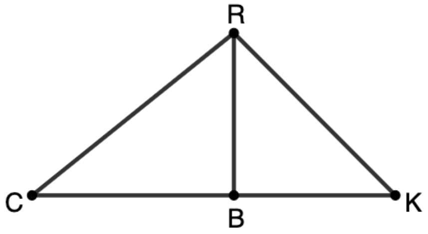
c. Find the measures of the segments and angles listed in the table. Round to the nearest thousandth.

Segment	Length	Angle	Measure
$\overline{MN}$		$\angle MNW$	
$\overline{CN}$		$\angle WMN$	
$\overline{CW}$		$\angle CWN$	
$\overline{KN}$		$\angle WCN$	
$\overline{MK}$		$\angle CKN$	

7.8.3 Guided Instruction: Solving Problems With Right Triangles

1. ΔRBK and ΔRBC are right triangles that share side $\overline{RB}$. $\angle RBK$ and $\angle RBC$ are both right angles, $m\angle BCR = 38.7°$, $m\angle RKC = 45°$, and $RB = 8$.

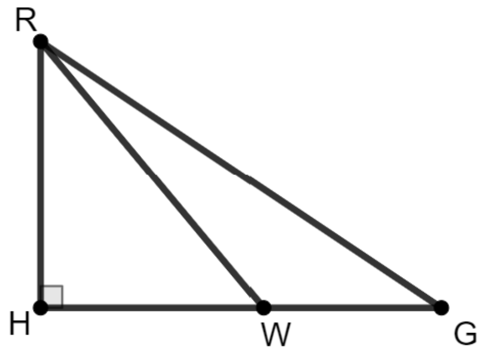
   a. Use the given information to complete the table. Whenever possible, give lengths as exact values. If necessary, round to the nearest hundredth.



Segment	Length
$\overline{RC}$	
$\overline{CB}$	
$\overline{RK}$	
$\overline{KB}$	
$\overline{CK}$	

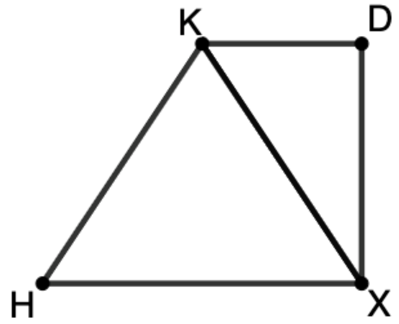
   b. What is the area of ΔCRK?

2. ΔRGH is shown, where $m\angle RHG = 90°$, $m\angle RWH = 50.2°$, $m\angle WRG = 16.5°$, $m\angle RGH = 33.7°$, and $RH = 12$.



What is the approximate perimeter of ΔRWG? Round your answer to the nearest hundredth.

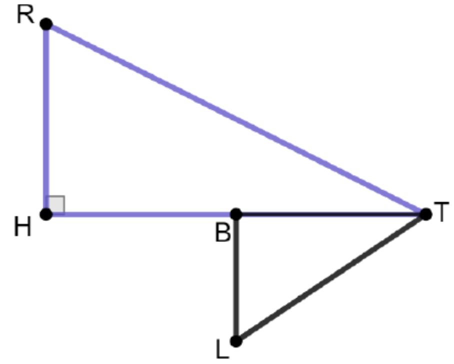
3. ΔKDX and ΔHKX share side $\overline{KX}$. ΔHKX is an isosceles triangle with base $\overline{HX}$ and ΔKDX is a right triangle with $m\angle KDX = 90°$.



If $HX = 13$, $DX = 9$, and $m\angle KXD = 34°$, what is the perimeter of quadrilateral $KDXH$? Round your answer to the nearest hundredth.

7.8.4 Guided Practice: Right Triangle Trigonometry and the Pythagorean Theorem

1. ΔRHT and ΔTBL are right triangles, where B is the midpoint of $\overline{HT}$, $m\angle RHT = m\angle TBL = 90°$, $m\angle RTH = 30°$, $BL = 3$, and $LT = 7$.

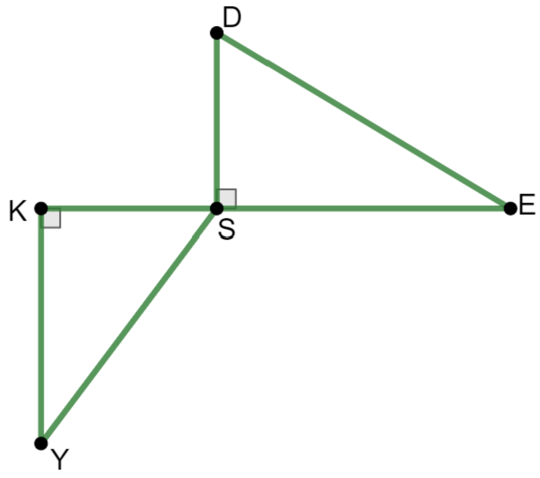


a. What is the length of $\overline{HT}$?

b. What is the area of ΔRHT?

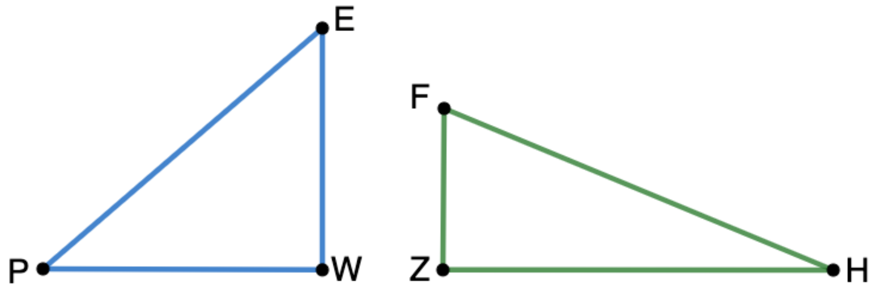
7.8.5 Your Turn!

1. In the diagram, ΔDSE and ΔYKS are right triangles that share the vertex S. The $m\angle DSE = m\angle YKS = 90°$, $m\angle DES = 31°$, $m\angle KSY = 53°$, $KE = 37$, and $DS = 10$.



Determine the length of $\overline{KY}$.

2. ΔPWE and ΔFZH are shown. In ΔPWE, $m\angle WEP = 49.3°$, $PE = 30.6$, and $\angle PWE$ is a right angle.



If $\angle FZH$ is a right angle, $\overline{FZ}$ is two-thirds the length of $\overline{EW}$, and $\overline{ZH}$ is 1.2 times longer than $\overline{PW}$, what is the length of $\overline{FH}$?

7.8.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 7, Lesson 9: Using Trigonometry in the Real World

Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Learning Target

☐ I can solve real-world problems using trigonometric ratios and the Pythagorean Theorem.